

Theorem. For real a, b , both roots of $p(z) = z^2 - az - b$ have modulus strictly below one if and only if

$$\boxed{a + b < 1, \quad b - a < 1, \quad b > -1.} \quad (\text{T})$$

This is the open triangle with vertices $(-2, -1), (2, -1), (0, 1)$. Stability here is a polynomial-root property; a stochastic-process statement requires additional assumptions.

1. Elementary proof. Write the roots, counted with multiplicity, as r, s . Then

$$r + s = a, \quad rs = -b, \quad p(1) = (1 - r)(1 - s) = 1 - a - b, \quad p(-1) = (1 + r)(1 + s) = 1 + a - b.$$

Thus (T) says exactly $p(1) > 0$, $p(-1) > 0$, $rs < 1$.

Necessity. If the roots are real and satisfy $|r|, |s| < 1$, both displayed products are positive and $rs \leq |r||s| < 1$. If they are nonreal, real coefficients give $s = \bar{r}$, so $p(1) = |1 - r|^2 > 0$, $p(-1) = |1 + r|^2 > 0$ and $rs = |r|^2 < 1$. Hence (T) holds in both cases.

Sufficiency. Suppose (T). For a nonreal conjugate pair, $|r|^2 = |s|^2 = rs = -b < 1$. For real roots, order $r \leq s$. If $s \geq 1$, positivity of $p(1)$ excludes equality and forces $r > 1$ as well, contradicting $rs < 1$. If $r \leq -1$, positivity of $p(-1)$ excludes equality and forces $s < -1$, again contradicting $rs < 1$. Therefore $-1 < r \leq s < 1$. Repeated roots are included. $\square$

2. Schur-step crosscheck. Using the programme's general Schur-step theorem, a polynomial is stable precisely when its constant coefficient has smaller modulus than its leading coefficient and its Schur transform is stable. At formal degree two,

$$p^*(z) = 1 - az - bz^2, \quad zq(z) = p(z) + bp^*(z) = (1 - b^2)z^2 - a(1 + b)z.$$

The first check is $|b| < 1$. Under it $1 \pm b > 0$, and the linear transform has root

$$\frac{a(1 + b)}{1 - b^2} = \frac{a}{1 - b}.$$

Its stability is equivalent to $|a| < 1 - b$. The two checks therefore say $-1 < b < 1$ and the first two inequalities in (T). Conversely, those two inequalities imply $2b < 2$, so (T) already implies $b < 1$. The Schur checks and (T) agree. This route uses the stated general theorem; the elementary proof above is independent of it.

Degeneracies and strict boundary. When $a^2 + 4b = 0$, the double root is $a/2$, stable exactly for $|a| < 2$. When $b = 0$, the roots are $0, a$, stable exactly for $|a| < 1$; the origin is a stable double-zero root. The edge $a + b = 1$ has a root at 1 ; $b - a = 1$ has a root at -1 . On $b = -1$, $-2 \leq a \leq 2$, the roots have modulus one, including the repeated endpoint roots. Every boundary point is excluded. The Schur cancellation was used only for $|b| < 1$, never at its endpoints.